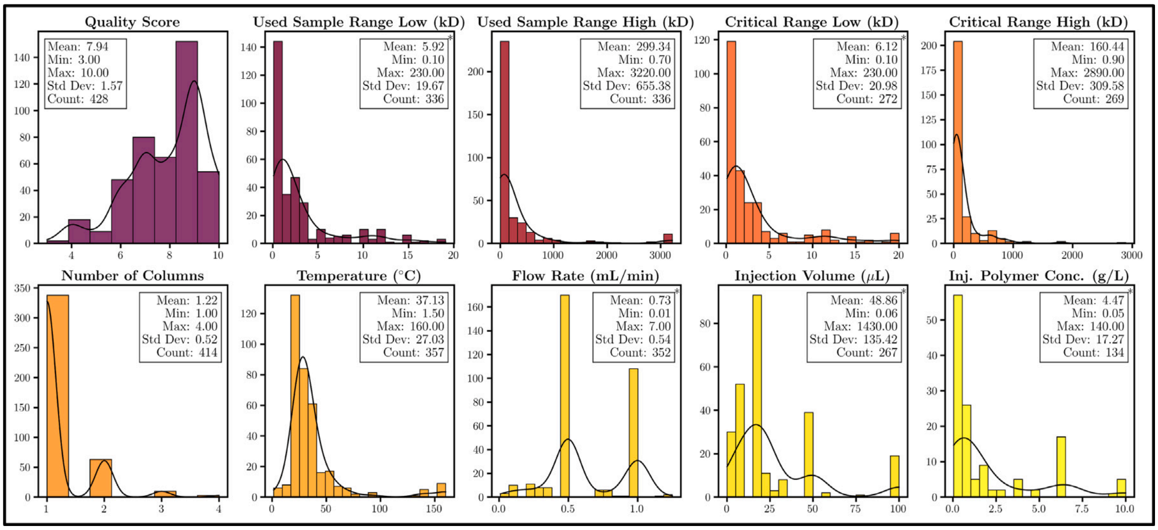


**Fig. 8.** Histograms of tracked numeric parameters with a box of key statistics. Vertical and horizontal axes display the count and value for each parameter. Black curves show the probability density function. Asterisks denote that the histogram displays only the lower 95th percentile.

current and future studies. By centralizing and streamlining access to critical chromatography conditions, this platform addresses long-standing challenges faced by researchers in accessing, sorting, and interpreting this information. Immediately, this resource offers polymer scientists a powerful tool for quickly identifying conditions that would otherwise require time-intensive literature searches. Over time, as the dataset expands and more researchers contribute their own findings, the platform will grow into a comprehensive resource, enabling advanced data analysis, cross-referencing, and ultimately accelerating the pace of discovery in the field. This database not only supports detailed studies of polymer properties but also opens up possibilities for exploring new critical conditions and optimizing experimental designs in an increasingly efficient manner.

Compared to existing resources, which often consist of static, printed compilations or small, standalone datasets, our database stands out as a dynamic, searchable, and continuously updated platform. It eliminates the need for repetitive data extraction from numerous studies by organizing conditions in a single, easily navigable repository. Unlike other resources, which tend to lose relevance over time, our platform is designed to grow with the field, incorporating both historical and newly submitted data. Additionally, as the field grows to include more critical data, we encourage researchers to use our platform to share their findings. The inclusion of submission capabilities empowers the community to participate actively in maintaining and expanding the dataset, fostering collaboration and reducing duplication of effort.

Beyond serving as a database, *PolyCrit* lays the groundwork for transformative applications in machine learning (ML) offering opportunities to uncover hidden patterns and predict critical conditions with greater accuracy. The current collection of polymers has already been converted to many descriptors we hope to use for ML purposes. One can elicit many numeric descriptors of a polymer including Hansen Solubility Parameters [15,16], as well as structural descriptors such as SMILES [17]. With sufficient data, ML algorithms could be trained to predict optimal critical chromatography conditions for new polymers based on structural or compositional similarities to previously characterized samples. This predictive capability would allow researchers to save time and resources by minimizing trial-and-error in laboratory experiments, offering initial conditions for experimentation that are likely to yield high-quality LCCC separations.

Furthermore, as the dataset grows, ML models could help identify underlying trends across different polymer types, such as correlations between branching patterns, $M_w$, and optimal separation conditions. Such insights could, in turn, guide the development of new polymer formulations and analytical methods, further advancing the field of liquid chromatography and broadening the applications of LCCC. To that end, we are currently working on developing a ML model to predict features of polymer critical conditions using our curated database. In the model, we plan on emphasizing principles of green chemistry to transition the field to environmentally sustainable solvents. We hope to bring this to our online platform to bolster the website's helpfulness as a tool for researchers. By integrating the power of data science and ML with experimental chromatography, *PolyCrit* has the potential to revolutionize both the study and application of LCCC.

## CRediT authorship contribution statement

**Brinton King Eldridge:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Dillon T.A. Baker:** Writing – review & editing, Investigation, Data curation. **Yongmei Wang:** Writing – review & editing, Supervision, Investigation, Data curation, Conceptualization.

## Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT to assist with rephrasing sentences and organizing key talking points. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgments

In preparation of this project, we would like to thank Nathan Dixon, Hunter Jackson, and Matthew Hosier, three University of Memphis Computer Science alums who helped build the initial framework for this website.

## Appendix A. Supplemental information

As to give proper credit to the papers present in *PolyCrit*, we cite them now [14,18–232].